

THE BUSINESS OF ENERGY TRANSITION (MBA292T.8)

FALL 2026 | THURSDAYS 4-6 PM at HAAS
Haas School of Business - Chou Hall Room N-570

I. Instructors

Cisco DeVries (bio here) Professional Faculty/ Lecturer UC Berkeley - Haas School of Business/ Goldman School of Public Policy - Former CEO of OhmConnect - Co-founder/ Former CEO of Renew Financial - US Department of Energy, US Department of Transportation, and City of Berkeley ciscodv@berkeley.edu Cisco DeVries' office hours • Thursday, 3-4pm at Haas (room 517 or nearby) for Sept and Oct • Mondays 4-5pm at GSPP (2607 Hearst living room) for Nov and Dec	Kate Gordon (bio here) Professional Faculty UC Berkeley - Haas School of Business - CEO, California Forward - Former Senior Advisor to U.S. Energy Secretary Jennifer Granholm - Former Senior Climate Advisor and Director of the Governor's Office of Planning and Research for Gov. Gavin Newsom khgordon@berkeley.edu To sign up for Kate Gordon's office hours (Thursdays between 3-4pm at Cafe Think) please use THIS LINK; email me for additional times.
Reader: Sam Willett, sam_willett@berkeley.edu <i>* Please direct any correspondence to both instructors and Reader.</i>	

II. Course Description

The energy transition is the largest capital reallocation in human history, and it is generating careers, companies, and investment opportunities at a pace no other sector can match. Using a combination of instructor-led discussion, readings, and guest speakers, this course prepares MBA students to compete for, and create, those opportunities.

The course is organized in three phases: the landscape, the toolkit, and the frontier. The **landscape** is an overview of our traditional energy economy, the work underway and progress in the energy transition, and a vision for what our energy system can look like in the coming decades.

In the **toolkit** phase, we work sector by sector through power, buildings, and transport. Each sector introduces a distinct set of technologies, business models and financial structures that range from large-scale project finance to behavioral economics. We examine the broader investment landscape—including carbon markets and blended public-private finance—and the specific ways clean energy investing differs from conventional finance. We also do a deep dive on some of the most important and topical issues, including corporate clean energy procurement strategies and the grid implications of load growth from data centers and advanced manufacturing.

The final **frontier** phase looks forward: asking what the energy system actually looks like as we adapt to a changing climate, emerging new technologies from long-duration storage to carbon removal, and a closing practitioner panel of Berkeley alumni leading organizations at the center of the transition. Guest speakers throughout are working practitioners, not observers.

No prior background in energy, engineering, or finance is required.

III. Requirements

Attendance and Participation

Students must attend every class session, complete all of the assigned readings and participate in class discussions. Please notify instructors in advance of any planned absences or if you require special accommodations. If it is absolutely necessary to miss a class, contact the class reader to provide you with the Zoom recording one time. To receive attendance credit for a missed class, you must write a memo on that class, including answers to discussion questions. Memos used to make up a missed class do NOT count towards the required 3 Class memos. Grades for all students will be based in part on class attendance and class participation.

This course features guest speakers who are leaders in the fields of climate change entrepreneurship, policymaking, finance, and investment. To ensure a vibrant discussion with these speakers, students are expected to come to class prepared with questions for our guest speakers in addition to responses to assigned study questions. Laptops are not allowed in class during guest presentations except as an approved accommodation.

Final Group Presentation Requirements

Students will form groups of **4 to 5 members** (subject to minor adjustments based on final enrollment) to develop and deliver a professional investment or business thesis tied to the energy transition.

Imagine your team is pitching to an **investment committee at a financial firm**. Your goal is to make a compelling case for investing in or developing a specific *category or sector* within the energy transition space. **Note:** Your presentation should focus on the broader sector or category opportunity, rather than pitching a single, specific company or startup.

Deliverables & Timing

- **Format:** A 12-slide presentation.
- **Duration:** 12 minutes for the presentation, followed by an 8-minute Q&A session with the committee (20 minutes total per group).
- **Participation:** All group members are expected to contribute equally to the conceptual development and participate equally during the oral presentation.

Your presentation must explicitly address the following five elements:

1. **Thesis Overview:** A concise, clear statement of your investment or business thesis.
2. **Market & Data Justification ("Why it Matters"):** Robust data and analytical evidence that supports the urgency and viability of your thesis.
3. **Value Proposition:** An analysis of both the economic returns and the broader societal/environmental benefits provided by this sector.
4. **Competitive Landscape:** An overview of existing or emerging business sectors, categories, or benchmark companies operating in this or similar spaces.
5. **Risk & Implementation Challenges:** A realistic assessment of the key technical, regulatory, financial, or societal hurdles associated with development and implementation.

Written Submission of 3 Memos

We require participation through both in-class comments and written memos. Students are required to submit **three written memos** over the course of the semester (one for each third of the term).

- **Length:** Maximum of 2 pages (shorter is perfectly fine).
- **Format:** Memos must be written in the professional style of an **investment thesis or business opportunity analysis**.
- **Scope:** The topics are broad to give you wide latitude on specific areas of interest. You are expected to integrate course materials and classroom discussions directly or indirectly—**outside research is not required**.
- **Deadlines:** Memos can be submitted at any time prior to their respective cut-off dates (detailed below in the *Summary of Deadlines*).
- **Grading:** **Up to 10 points** for all memos that fulfill the baseline prompt requirements, adhere to the length constraint, and demonstrate critical thought. **Bonus points** for memos that show

outstanding analysis, unique synthesis of course concepts, or a highly compelling and unique approach.

Two-Page Memo Topics

- **Memo 1: Analyzing the Transition Landscape.** Select a single slide from Nathaniel Bullard's 2026 slideshow (available at nathanielbullard.com/presentations). Address the following in your analysis:
 - What critical story does this slide tell us about the current state of the energy transition?
 - What key nuances, data points, or perspectives do you think this slide misses?
 - Suggest a startup business thesis that leverages the trend shown in the slide, drawing on class readings and discussions to support your case.
- **Memo 2: The Role of Policy and Regulations in the Energy Transition sector.** Select a single sector within the energy transition landscape (e.g., building retrofits, low-carbon fuels, EV charging infrastructure, or grid storage). Address the following in your analysis:
 - Identify a significant policy or regulatory change that you think is likely (or at least possible) to occur over the next 2–3 years.
 - Formulate a professional memo evaluating the specific business opportunities and risks that will emerge as a result of this shifting regulatory landscape.
- **Memo 3: Emerging Technology Integration**
 - **Topic:** AI as a problem and/or tool for solving energy transition challenges.
 - **Prompt:** Identify a specific bottleneck in the energy transition (e.g., grid management, battery supply chains, or renewable forecasting). Formulate a business or investment thesis evaluating how, either:
 - AI can be utilized to solve bottlenecks that are created.
 - Energy transition opportunities will emerge as we address the bottlenecks brought on by the growth of AI and/or the resources needed to run it.

Summary of Grading (100 point scale)

10% Attendance (10 points)

20% In Class Participation (20 points)

- Students are expected to attend every class; if you have to miss a class you will need to write an additional memo focused on the topic and reading of that missed class, on top of the 3 memos already required.

20% 3 Written Class Memos (20 points)

50% Final Project -- Group Presentation (50 points)

(Up to 5 bonus points can be earned)

Summary of Deadlines

Written Memos:

- **Wednesday, September 16, 2026** - Memo 1 is due
- **Wednesday, October 14, 2026** - Memo 2 is due
- **Wednesday, November 4** - Memo 3 is due

Final Project:

- **Thursday, November 4:** Notify instructors of team formation and tentative topic
- **Thursday, Nov. 12:** Sign up for your preferred date (Dec 3 or 10)
- **Thursday, Dec 3:** All final slide decks due prior to the start of class
- **Thursday, Dec. 3:** Final Presentations
- **Thursday, Dec. 10:** Final Presentations

Prerequisites and Class Participants

No prior background in energy, engineering, or finance is required.

Class Format & Materials

Some lecture, discussion, guest speakers. Will include a mix of written memos, in-class debates, and a final group presentation. Students are required to read / watch / listen to the items listed under "Materials." Items listed as "*Recommended*" are optional, and included for students who are interested in taking a deeper dive into particular subjects. In some cases, recommended materials may relate directly to study questions, offering information not fully captured in the required materials.

Taping Policy

Classes will be taped with access for enrolled students for a limited time after each class. Duplicating, disseminating or taking screenshots is strictly prohibited. See absence make-up policy above.

AI Rules**Some AI Use Permitted (with disclosure)**

With some restrictions, students are permitted to use generative AI tools for in this course. Learning to appropriately use AI as a tool is becoming an important component of business education. The key principle underlying this policy is that students remain responsible for the integrity and originality of their work. Broad copying of generative AI is plagiaristic and is prohibited. If unauthorized AI use is suspected, an examination of the work and/or a separate oral examination on the content may be required.

All memos and presentations must include a clear statement (on the front page or slide) that details how AI was used.

Permitted uses for AI tools include:

- Using AI to brainstorm topics or generate outlines (you must rephrase and develop these ideas in your own words).

- Using AI for writing assistance, such as correcting grammar, spelling, or suggesting synonyms.
- Using AI for translation, for the purpose of consultation.
- Using AI to summarize information (you must verify the source materials and content yourself).

IV. Class Overview and Reading List

Week 1:

So what is the energy transition? And where are we on it?

August 27, 2026

Summary

This introductory week establishes the foundational framework for understanding the global energy transition. We will explore the core definition of the transition, framing it as:

- A massive reallocation of capital from fuel-based operating expenses (OPEX) to technology-based capital investments (CAPEX);
- A dynamic shift from a primarily centralized system to a significantly decentralized one; and
- A move from carbon-based energy to a carbon-free energy system.

What's more, we are making this transition more quickly than any energy transition in human history. We will touch on the basics of electricity systems and fundamental financial concepts necessary for the course. The session will also establish a lens that runs throughout the entire course: energy is a regulated industry, not a free market, and policy creates and destroys business models as surely as technology does.

Speakers: Kate Gordon, Cisco DeVries

Materials

- Nathaniel Bullard, *Decarbonization: Parameters, Dollars and Sense, Electrons Photons Molecules* (presentation deck) (<https://www.nathanielbullard.com/presentations/>)
- BNEF: *New Energy Outlook 2026* (PDF) slides 1-6: <https://about.bnef.com/insights/clean-energy/new-energy-outlook/#download-report-summary>
- RMI Insight: *The Great Reallocation* (<https://rmi.org/resources/the-great-reallocation/>)
- Vaclav Smil: *ENERGY AND CIVILIZATION: A HISTORY. Chapter 1: Energy and Society* (available online at <https://ieeexplore-ieee-org.libproxy.berkeley.edu/document/7986041>)
- **Optional/extra:** Kate Gordon, *Managing the Energy Transition: A Place-Based Approach* (Carnegie California, 2024), available at

<https://carnegieendowment.org/research/2024/08/managing-the-energy-transition-a-place-based-approach>

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Week 2:

The legacy (carbon-based) energy economy

Sept 3, 2026

Summary This session focuses on the traditional, carbon-based economy. We will analyze the foundational systems, structures, and economic dependencies that have historically driven fossil fuel reliance, setting a baseline to contrast against the low-carbon transition discussed in subsequent weeks.

Speakers Bill Bartling, fmr Chief Deputy, CalGEM (confirmed)

Materials

- Our World in Data, *Fossil Fuels: basic overview of history and current production/consumption*, available at <https://ourworldindata.org/fossil-fuels>
- Oxford Institute for Energy Studies, *Unpacking the Hormuz Crisis: Implications for energy markets and the energy transition* (Introduction and chapters by Fyfe, Bordoff & O'Sullivan, Bouchoev), available at <https://www.oxfordenergy.org/wpcms/wp-content/uploads/2026/05/OEF-149.pdf>
- Samantha Gross, *Why Are Fossil Fuels So Hard to Quit?* (Brookings Institution June 2020). Available at: <https://www.brookings.edu/articles/why-are-fossil-fuels-so-hard-to-quit/>

Week 3:

Visions from the future, a low carbon economy

September 10, 2026

Summary What if it works? This session explores the potential for a successful transition to a low-carbon economy. We will focus on the vision of 2050 and, more importantly, on the *path* to 2050 – breaking down the massive deployment of zero-carbon power, electrified transport networks, and dynamic grids required to structurally replace fossil fuels. We will explore the practical, real-world solutions necessary to drive this transition forward without compromising economic growth.

Speakers Saul Griffith, Rewiring America (confirmed, virtual)

Materials

- Saul Griffith, *ELECTRIFY: AN OPTIMIST'S PLAYBOOK FOR OUR CLEAN ENERGY FUTURE* (Chapters 1-8)
- <https://www.iea.org/reports/net-zero-roadmap-a-global-pathway-to-keep-the-15-c-goal-in-reach>
- Arup Digital Resource: *2050 Scenarios: Four Plausible Futures*
<https://www.arup.com/en-us/insights/2050-scenarios-four-plausible-futures/>

Week 4:

Toolkit: The transition to low carbon power at the utility scale

September 17, 2026

Summary This session introduces the regulatory and structural toolkits governing utility-scale power systems. We will explore how traditional utilities operate, manage macro-level infrastructure, and respond to state-level mandates. To understand the practical execution of these systems, we will contrast a major regulatory triumph—the deployment of Renewable Portfolio Standards (RPS) that successfully scaled utility solar and wind—against a critical structural failure, analyzing how deferred grid maintenance and poor risk management can lead to catastrophic infrastructure and service collapses.

Speakers Elliot Mainzer (confirmed - virtual)

Materials

- New York Times interactive: *America's power grid* (overview of current infrastructure, increasing energy demand)
https://www.nytimes.com/interactive/2026/04/27/opinion/electricity-power-grid-infrastructure.html?unlocked_article_code=1.eFA.8DMF.P9gfUk6o80O1&smid=nytcore-ios-share
- Lawrence Berkeley National Laboratory, U.S. State Renewables Portfolio & Clean Electricity Standards: 2024 Status Update, available at
https://eta-publications.lbl.gov/sites/default/files/lbnl_rps_ces_status_report_2024_editio_n.pdf
- Katherine Blunt, *CALIFORNIA BURNING: THE FALL OF PACIFIC GAS AND ELECTRIC--AND WHAT IT MEANS FOR AMERICA'S POWER GRID* (2022), Chapters 1-5, available online at
<https://research-ebsco-com.libproxy.berkeley.edu/c/p54vgy/search/details/x74yszdq6z?request-context=plink&db=nlebk>
- Meredith Fowlie, *This Public Power Movement is Raising a Billion Dollar Question* (Energy Institute at Haas blog post March 9, 2026), available at
<https://energyathaas.wordpress.com/2026/03/09/this-public-power-movement-is-raising-a-billion-dollar-question/>

Week 5
Toolkit: The transition at the grid edge

September 24, 2026

Summary

This session explores the transition at the "grid edge," analyzing the technical and financial toolkits governing the growth of decentralized, local energy systems. We will examine the critical structural shift from a centralized grid to a dynamic network of active "grid participants" utilizing Distributed Energy Resources (DERs) to generate, store, and manage power. Through real-world case studies, the lecture will break down how flexible generation and demand are aggregated and monetized. We will walk through the specific hardware integrations, software optimization, and project financing structures required to construct and operate "virtual power plants" (VPPs) at scale.

Speakers: Cisco DeVries and Peter Gage

Materials

- *Pathways to Commercial Liftoff: Virtual Power Plants* (US Department of Energy). Executive Summary.
- *Realizing the Virtual Power Plant Opportunity* (The Brattle Group, 2023)
- FERC 2222 Primer:
<https://www.ferc.gov/ferc-order-no-2222-explainer-facilitating-participation-electricity-markets-distributed-energy>
- <https://www.youtube.com/watch?v=HOUTzlgDfT8>
- <https://www.youtube.com/watch?v=oUbC6eMMKds>
- <https://www.youtube.com/watch?v=wt5FOX5L7G0>

Week 6:
Toolkit: Buildings

Oct 1

Summary This session unpacks the toolkits driving decarbonization in the built environment, separating the challenge into two distinct categories: optimizing new construction and retrofitting existing real estate. We will analyze the growing need for building materials, embodied carbon, and the operational economics of electrification (heat pumps, induction, and thermal storage).

Speakers Panama Bartholomy, Building Decarbonization Coalition (confirmed, in person)

Materials

- Meredith Fowlie, *Why are we sinking billions into a gas system we plan to leave behind?* (Energy Institute at Haas blog post, July 20 2026).
<https://energyathaas.wordpress.com/2026/07/20/why-are-we-sinking-billions-into-a-gas-system-we-plan-to-leave-behind/>
- <https://www.volts.wtf/p/how-to-phase-out-residential-gas>
- <https://rmi.org/resources/the-economics-of-electrifying-buildings-residential-new-construction/>
- <https://carbonleadershipforum.org/2025-embodied-carbon-reductions-built-project-case-study-collection/>

Week 7:

Toolkit: Transportation

Oct 8

Summary This session analyzes the commercial landscape of the clean transportation transition, focusing on the market forces and business models scaling zero-emission mobility. We will evaluate the massive unit economics, capital expenditure (CAPEX) hurdles, and supply chain bottlenecks involved in electrifying commercial fleets and deploying public charging infrastructure. The lecture will explore the market opportunities in heavy-duty transit and fleet commercialization, while examining how regional infrastructure megaprojects (like High-Speed Rail) alter long-term real estate values and economic corridors. Finally, we will address how state-level mandates (such as California's 2035 zero-emission vehicle targets) act as a secondary catalyst—shaping market demand, creating regulatory credit markets, and forcing corporate strategy shifts.

Speakers Warren Logan (invited)

Materials TBD

Week 8:

Toolkit FINANCE

Oct 15

The energy transition is, at its core, a financing problem. The structural shift from fuel-based operating expenses to technology-based capital investment is an enormous financing challenge

by itself. Add to that a dramatic move from centralized to decentralized energy, and we essentially need to reinvent the entire financing system for energy. Then, on top of that, we are trying to transform transportation in a way that requires everything from charging infrastructure to large public private partnerships.

This session builds a working framework for how clean energy projects get financed. Scott Jacobs, co-founder of Generate Capital, will examine how private project finance underwrites clean energy infrastructure at scale — walking through the capital stack, return expectations, and how asset-backed structures have unlocked institutional capital for the sector.

Speakers: Scott Jacobs, former Generate Capital (confirmed, in person)

Materials:

- Electrify: An Optimist's Playbook for Our Clean Energy Future by Saul Griffith (Chapters 9-12)
- Lara Pierpoint, [Mobilizing Catalytic Capital in Climate Infrastructure](#), (April 2023), Insiders Guide to Energy [Podcast]
- Baysa Naran et al., [Global Landscape of Climate Finance: A Decade of Data](#), Climate Policy Initiative (Oct. 2022).
 - *Content: Brief summary of global climate finance trends since 2011 and finance needs for the next decade. Full report available [here](#).*
- Jeff St. John, [Small Island Nations Suffer Under Broken and 'Cruel' Climate Finance System](#), Canary Media (Nov. 3, 2021)

Week 9:

Toolkit: AI and Data Centers

Oct 22

Summary This session examines the massive commercial bottleneck at the intersection of artificial intelligence, cloud computing, and grid capacity. We will analyze the sheer unit economics and infrastructure demands of next-generation data centers, treating energy availability not just as an operational cost, but as the primary limiting factor for tech sector scaling and valuation. The class will evaluate the dual-nature of this challenge: exploring how big tech companies deploy venture capital and corporate procurement to secure massive power pipelines, while conversely analyzing the business models of climate-tech companies utilizing AI to optimize grid transmission, speed up project siting, and reduce macro-level energy waste.

Speakers Carla Peterman, PG&E (confirmed)

Materials

- Data centers and energy - impact on affordability/load management
- AI as a tool in the transition (i.e. companies using AI to speed the energy transition)
- Tom Wheeler, *Data center backlash signals a fight over AI power* (Brookings Institution July 7, 2026), available at <https://www.brookings.edu/articles/data-center-backlash-signals-a-fight-over-ai-power/>
- <https://www.mckinsey.com/industries/technology-media-and-telecommunications/our-insights/the-cost-of-compute-a-7-trillion-dollar-race-to-scale-data-centers>
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Week 10:

The Politics of Energy Transition

Oct 29

Summary This session analyzes how distinct political landscapes and regulatory architectures dictate corporate strategy, project development timelines, and investment risk. We will open with a real-time business debrief on the immediate economic implications of the recent midterm election results. The core of the lecture will evaluate the ultimate clean energy case study: California vs. Texas. California and Texas are the two largest energy economies in the nation, but they represent the ultimate clash of business environments: California's highly regulated, mandate-driven, high-CAPEX model versus Texas's deregulated, market-clearing, rapid-deployment model.

Speakers Jennifer M Granholm, former US Energy Secretary (confirmed)

Materials: Will be assigned after the election

Week 11:

Consumers & Behavioral Economics

Nov 5

Summary

This session explores the intersection of behavioral economics and public utility engagement, focusing on the critical role of the consumer in the modern energy economy. The first hour features a deep dive on the importance of engaging consumers as a vital tool for the energy transition, highlighting strategies that leverage everything from behavioral nudges and normative social influence to shift demand. In the second hour, we will dive into a practical case study and open discussion with OhmConnect, a pioneering company that operated at the

cutting edge of consumer-facing grid response, to analyze real-world applications, challenges, and successes in mobilizing energy users.

Speakers

Matt Duesterberg, former energy trader; Cadir Lee, former Zynga—both founders of OhmConnect (confirmed, in person)

Materials

- OhmConnect HBS case
- *Nudge: Improving Decisions About Health, Wealth, and Happiness* (co-authored by Nobel laureate Richard H. Thaler and legal scholar Cass R. Sunstein) Selected chapters
- OPOWER: Increasing Energy Efficiency through Normative Influence (HBS Case 911-016)
- https://www.youtube.com/watch?v=HqR_tqRZmV0

Week 12:

Toolkit: Corporate Forms & Financial Institutions

Nov 12

Summary This session explores the shifting corporate and financial architecture driving the low-carbon transition, analyzing why major corporations have emerged as primary drivers of renewable energy demand. Moving past surface-level sustainability narratives, we will evaluate the strict business logic behind advanced corporate procurement strategies—contrasting physical versus virtual Power Purchase Agreements (PPAs), Renewable Energy Certificates (RECs), and the operational hurdles of 24/7 carbon-free energy matching. The lecture will dissect the financial and legal realities of Scope 1, 2, and 3 emissions accounting, equipping students with the frameworks needed to critically assess corporate net-zero commitments and separate credible corporate strategy from performative optics.

Speakers Susan MacCormac, Morrison & Foerster LLP (confirmed, in person)

Materials

- Rebecca Henderson, *To Save the Climate, We Have to Reimagine Capitalism*, TED (Nov. 2020). [Video, 8 min]
- Richard Kaufman & Marc Roston, [Chapter 5: Fixing the Plumbing: Asset Management, Clean Energy Technology, and the Valley of Death](#), *SETTLING CLIMATE ACCOUNTS: NAVIGATING THE ROAD TO NET ZERO* (2021).
- Mariana Mazzucato, *Introduction: Making versus Taking*, [The Value of Everything](#) ([video presentation](#) 2018).
- Susan Mac Cormac, [A Startup Cheat Sheet for the Baffling World of Corporate Structures](#),

Medium (Jan. 2017).

Week 13:

Frontier Panel

with Gabriel as host (alumni or incubators)

Nov 19

Summary We are moving away from the technologies that are commercially viable today and entering the multi-billion dollar world of venture bets, deep-tech underwriting, and future market creation. This session explores the commercial frontier of climate technology, focusing on the capital deployment frameworks required to scale uncommercialized, high-CAPEX technologies. Moving past theoretical science, we will analyze the business architecture backing Direct Air Capture (DAC), green hydrogen, Small Modular Reactors (SMRs)/fusion, and geoengineering. The lecture will dissect how venture capital firms underwrite deep-tech risk, how corporations utilize Advanced Market Commitments (AMCs) to jumpstart frontier supply chains, and how project developers manage the intense "valley of death" between laboratory breakthroughs and commercial bankability.

Speakers Gabriel Kra, Managing Director of Prelude Ventures, will speak and help moderate a pitch session from new deep tech start ups

Weeks 14-15:

Group Presentations

Dec 3 & Dec 10

Summary Group Presentations